Dale Automation

Training - Prover Maintenance & Operation

Inspection Details

Report #: Dale Automation 2801

Location:

Inspection Date: 23-Oct-2026 08:39 PM

Inspector Name: Sean Frost

Asset Details

Asset Category : 8000 LPH Proover

Asset Name : Blue Ribbon Middleburg

Asset # : DSS 403-4

Meter Reading: 0 Hours

Model/Year: Dale Bakermation

1.1 Loader Operation

SN	Inspection Item	Note	Good	Repair	Replace	NA
1	Fortress Lockout System	- Fortress lockout system & Run-inhibit protocol. - E-Stop. - Key entry system (Loader, Main Prover, Unloader). - Load Part Batch function. - Manual Homing function.	✓	○	○	○
2	Magnet Conveyor & Safety Cage System	- Magnet conveyor operation. - Magnet cage safety switch lockout protocol.	✓	○	○	○
3	Cross Conveyor Batching Sensors	- "Batch Made", "Push Position" & "Extended Batch" Proxies.	✓	○	○	○
4	Loader Pusher Arm Position Proxies	- Pusher Rear Home, Pusher Ultimate Rear Home, Pusher Mid, Pusher Ultimate Front Proxies.	✓	○	○	○
5	Loader Arm Pneumatic Arm Position Proxies	- Loader Pusher Arm Raised proxy. - Loader Pusher Arm Lowered proxy.	✓	○	○	○
6	Pusher & Cross Conveyor Motor Isolators	- Loader Pusher Isolator - Loader Cross Conveyor Isolator	✓	○	○	○
7	Loader In-Place Proxies	- Control Side Loader In-Place Proxy - Non-Control Side Loader In-Place Proxy	✓	○	○	○
8	Infeed Under/Over Push & Cross Finger Photocells	- Over/Under push photocell protocol.	✓	○	○	○
9	Safety Mechanisms (Fortress Lockout, E-Stops, Pull Chords)		✓	○	○	○
10	Finger Tilt Proxies	- Finger tilt triggered condition & safe reset procedure.	✓	○	○	○
11	Pulse Sprocket	- Pusher Driveshaft pulse sprocket operation	✓	○	○	○

SN	Inspection Item	Note	Pass	Fail	NA
12	Swing Tilt Proxies	- Swing Tilt proxy triggered condition & reset procedure explain	✓	○	○

1. Fortress Lockout System



2. Magnet Conveyor & Safety Cage System



3. Cross Conveyor Batching Sensors





4. Loader Pusher Arm Position Proxies



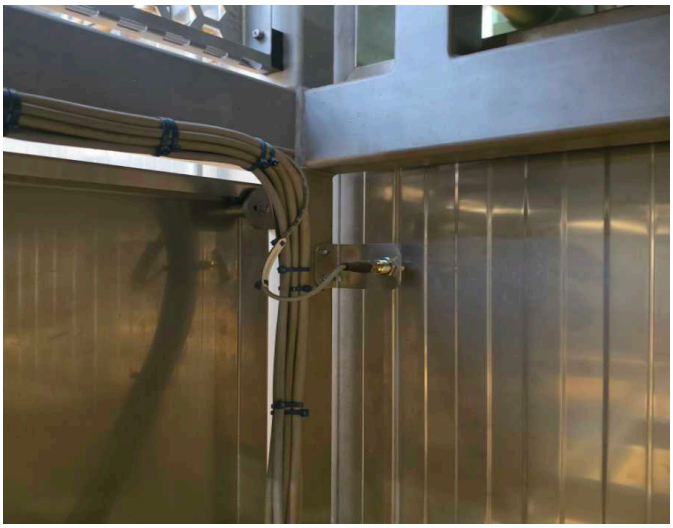
5. Loader Arm Pneumatic Arm Position Proxies



6. Pusher & Cross Conveyor Motor Isolators



7. Loader In-Place Proxies

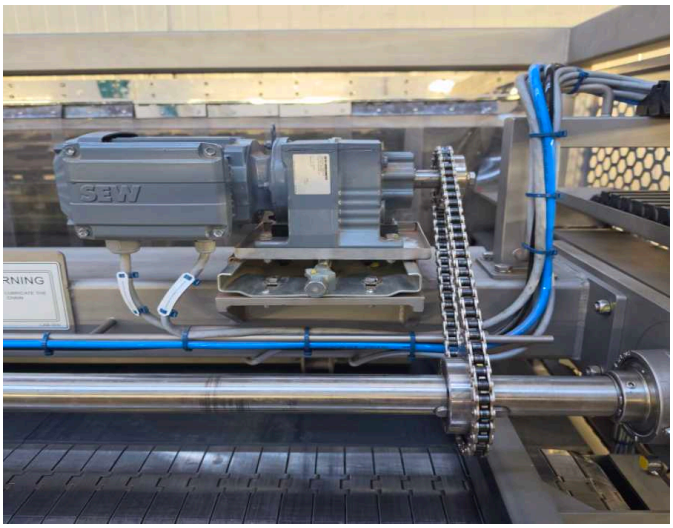




1.2 Loader Maintenance

SN	Inspection Item	Note	Good	Repair	Replace	NA
13	Gearmotor/Drive Mechanism Assembly	- Pusher chain tensioned. Sprocket tooth inspection. - Cross conveyor bearings greased. - Gearbox oil samples/6-12 monthly oil change. - Noise/vibration monitoring.	✓	○	○	○
14	Bearings, Rollers, Rack & Pinion, Drive Coupling	- Grease bearings monthly. - Routine clean & re-grease of rack & pinion. - Roller inspection - Main drive coupling inspection.	✓	○	○	○
15	Pneumatic System	- Clean ASU filter/drain condensate. - Verify setpoint is maintained (6Bar). - Leak detection.	✓	○	○	○
16	Cross Conveyor	- Slat belt & wearstrip inspection. - Sprocket & roller inspection. - Bearing inspection/re-grease.	✓	○	○	○
17	Electrical Checks	- Inspect batch proxies, terminal position proxies, pusher position proxies, encoder cables, cable igus.	✓	○	○	○
18	Safety Mechanism Tests	Routine test loader panel E-Stop, terminal position proxies, extended batch proxy.	✓	○	○	○

13. Gearmotor/Drive Mechanism Assembly

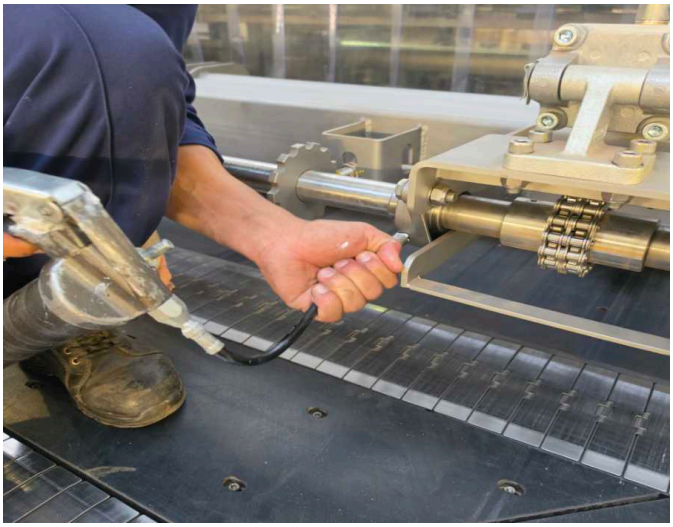




14. Bearings, Rollers, Rack & Pinion, Drive Coupling

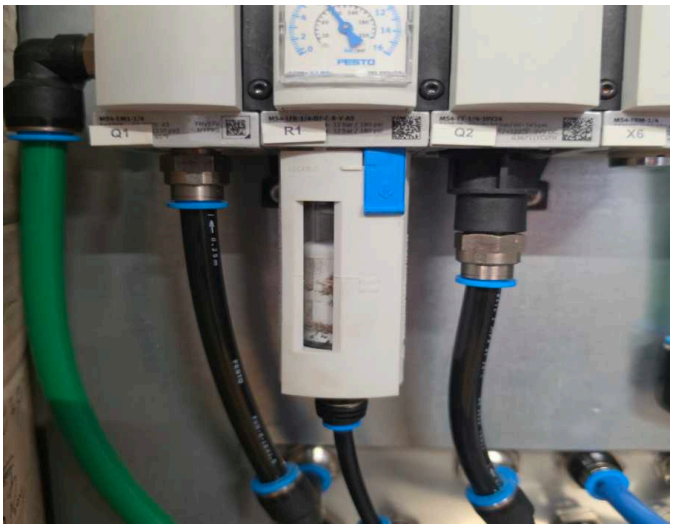






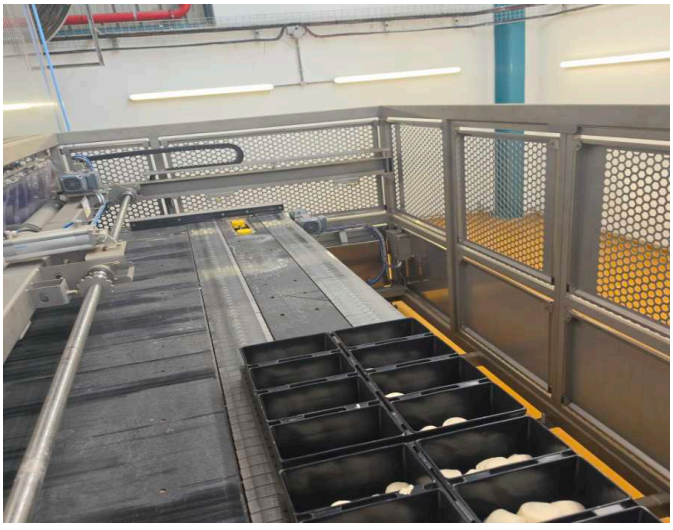
15. Pneumatic System



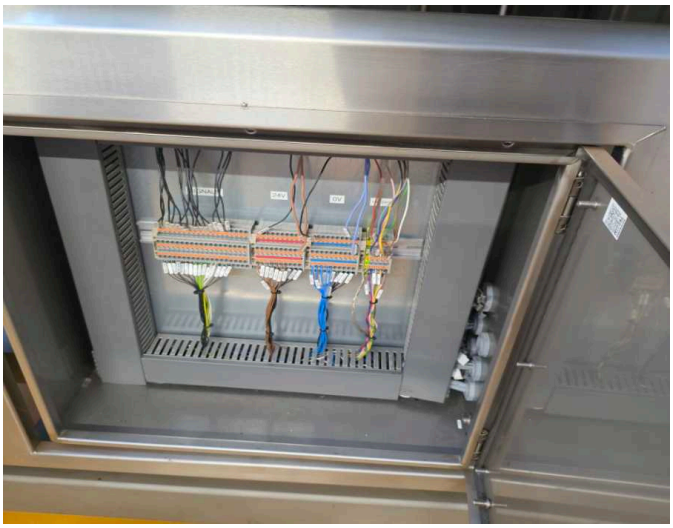




16. Cross Conveyor



17. Electrical Checks



18. Safety Mechanism Tests







2.1 Main Prover Operation

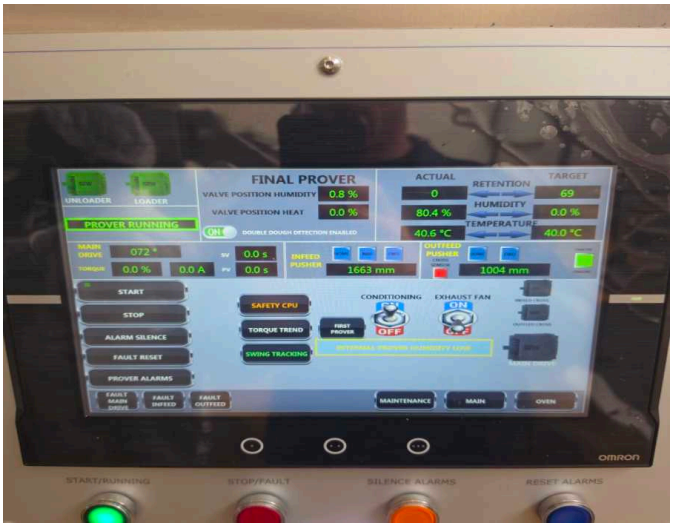
SN	Inspection Item	Note	Good	Repair	Replace	NA

19	Main Drive Gearbox	- Start/Stop Main Drive. - Start/Stop Air-Conditioning (Steam & circulation fan start-up). - Initiate Exhaust sequence. - Retention time control. - Temperature/Humidity Control. - Alarm silence/reset. - Push button control - Swing tracking. - Prover motor blocks. - Double dough detection On/Off. - Prover overview page. - Damper manual control.	✓	○	○	○
20	Main Drive Motor Brake	- Explanation of general alarms and alarm conditions. - Explanation/demonstration of safe recovery & reset procedures. - Alarms history page. - Safety CPU status page.	✓	○	○	○
21	Chain Tensioners	- Correct operation of tensioner system, compression set-points to be adhered to and monitored.	✓	○	○	○
22	Chain Oiler System	- Oiling sequence & system operation (charging of injectors) - Manually force pump or manually start oiling cycle.	✓	○	○	○
23	Airconditioning System	- Explanation of air circulation system by means of a Supply & return fan. - Explanation of Exhaust, Coil & Fresh Air intake dampers and their operational sequences.	✓	○	○	○
24	Steam System	Explanation of steam system, temperature line & location, humidity line & location, steam coil, steam injection through steam lance, modulating valves, pressure regulating valves and required setpoints.	✓	○	○	○

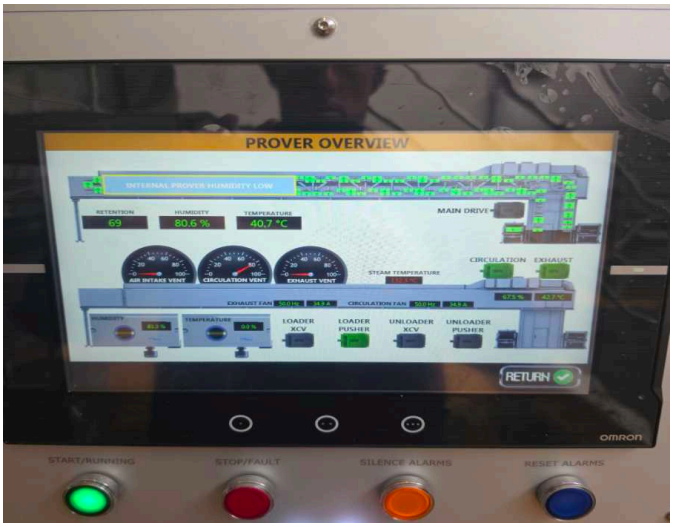
SN	Inspection Item	Note	Yes	No	NA
25	Maintenance Screens	- Chain oiler page: Manual oil cycle/Manual oiler force/adjustment of injector charge frequency & oiling frequency. - Conditioning PID & torque trending. - Loader/Unloader manual control.	✓	○	○

SN	Inspection Item	Note	Pass	Fail	NA
26	Manual Pendant	- Operation of manual pendant main drive control. - General situations to use manual control/driving to "safe" positions.	✓	○	○
27	Encoder Assembly & Back-Up Encoder	- Position control via encoder assembly, coupled via timing belt to idle sprocket. - Back-up encoder tabs & proxies, indication lights. - Manual pendant control & operation.	✓	○	○

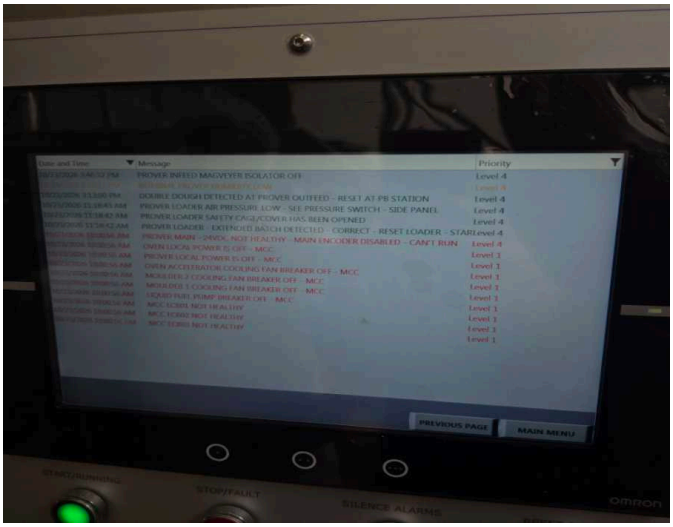
19. Main Drive Gearbox

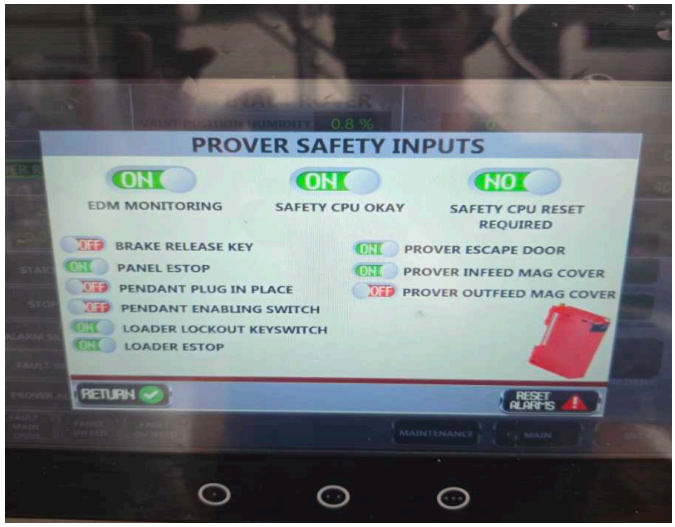






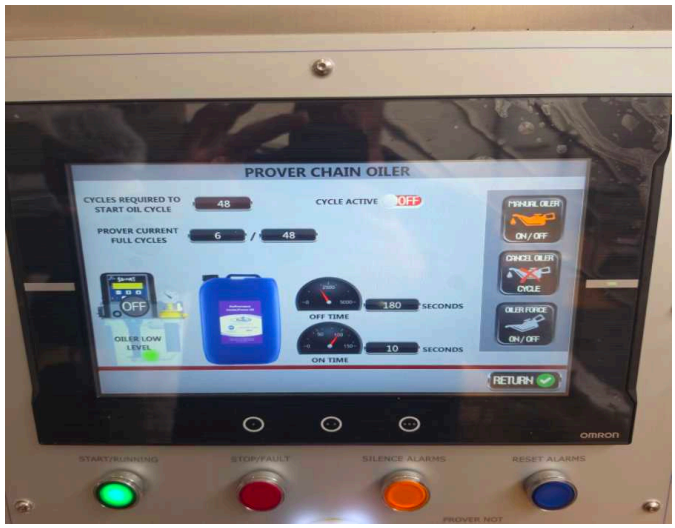
20. Main Drive Motor Brake



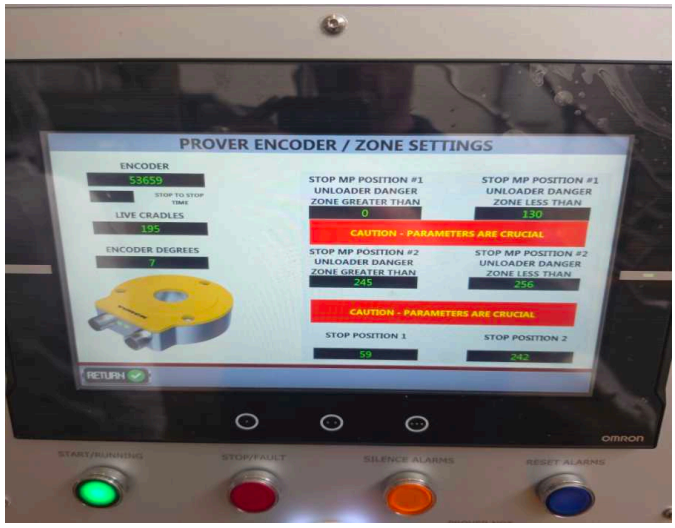


21. Maintenance Screens

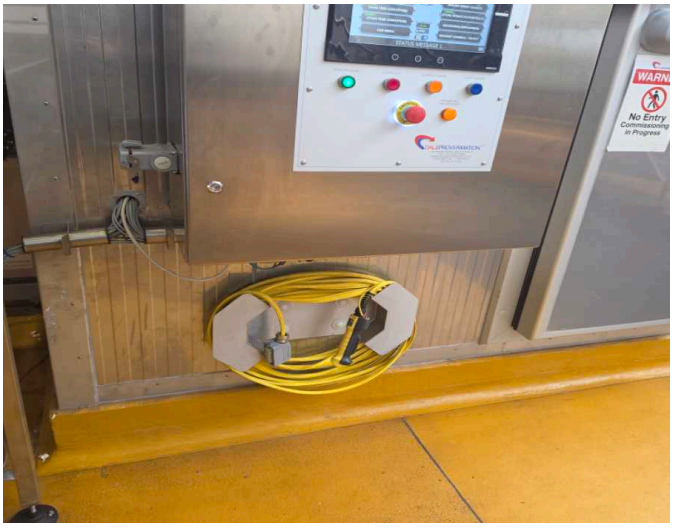








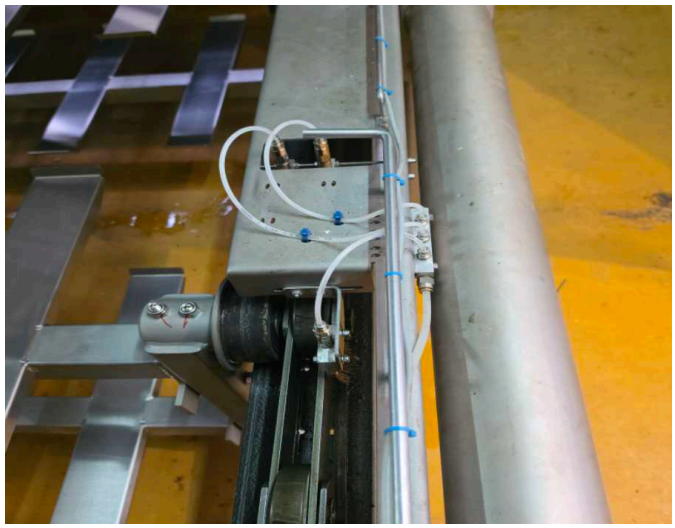
23. Manual Pendant





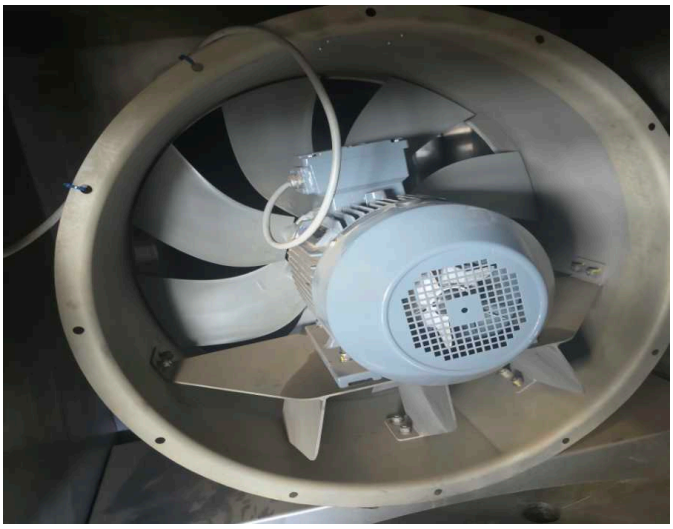
24. Chain Oiler System







25. Airconditioning System

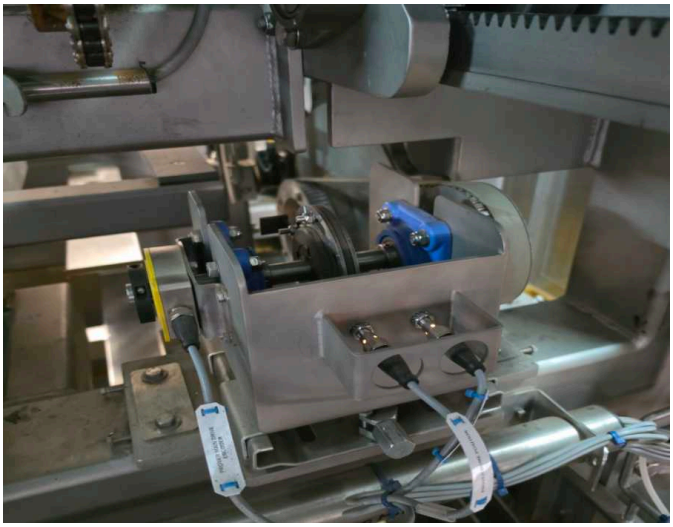








27. Encoder Assembly & Back-Up Encoder



2.2 Main Prover Maintenance

SN	Inspection Item	Note	Good	Repair	Replace	NA
28	Main Drive Gearbox/Motor Brake	- Safety Lockout Procedures when entering prover to perform work - Lockout of Main Drive/Unloader/Loader Fortress Lockout - Oil sampling/6-12 monthly oil change. - Monitor for leaks/noise/vibration. - Mounting/torque arm inspection for tightness. - Brake engaging/releasing (Relay inspection). - Routine verification of brake gap.	✓	○	○	○
29	Main Drive Bearings	- Inspection for condensate in grease/grease breakdown. - Inspect seals. - Monitor bearings for noise/vibration. - Routine top-up of grease.	✓	○	○	○

30	Encoder Assembly	- Inspect timing belt tension, alignment/signs of wear. - Inspect Encoder & Puk tightness. - Verify feedbacks & values for full rotation on HMI.	✓	○	○	○
31	Swing Inspection	- Swing inspection for straightness/corrosion. - Roller inspection: Turning freely, roller flanges for wear, bearings & seals, swing pins are recieveing sufficient oil.	✓	○	○	○
32	Chain, Sprocket & Tensioner Inspection	- Routine measurement of set number of links to monitor chain stretch. - Chain condition: signs of wear, sufficient lubrication on links/pins/rollers. - Sprocket teeth for signs of wear. - Regularly grease tensioner slides PF FG 2. Verify compression at setpoint.	✓	○	○	○
33	Oiler Pump & Brushes	- Pump Oil level. - Oiler brush condition. - Pump operation test (Manually force from maintenance screen). - Inspect injectors, oiler blocks & pipework for leaks.	✓	○	○	○
34	Steam Train	- Cleaning of strainers on main & condensate line. - Cleaning of DP27 PRV filter. - Verify design setpoints maintained during production. - Leak detection.	✓	○	○	○
35	Safety Circuit Test & Verification	Routine test of all E-Stops, door lockouts, finger/swing tilt proxies, finger clear photocells.	✓	○	○	○
36	Air-conditioning System	- Inspect ducting for corrosion/leaks/debris. - Greasing of circulation fan bearings (15000hrs/SRI-2). - Monitor circulation fans for excessive noise/vibration.	✓	○	○	○

28. Main Drive Gearbox/Motor Brake



29. Main Drive Bearings



31. Swing Inspection



32. Chain, Sprocket & Tensioner Inspection



34. Steam Train









3.1 Prover Unloader Operation

SN	Inspection Item	Note	Good	Repair	Replace	NA

37	Unloader Rear Home & Front Push position Proxies	- Pusher Rear Home, Pusher Ultimate Front/Offload Position Proxies.	✓	○	○	○

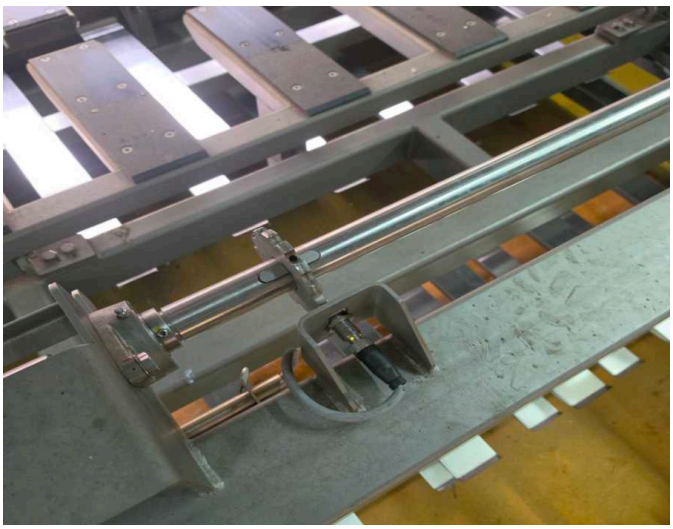
SN	Inspection Item	Note	Pass	Fail	NA
38	Pulse Sprocket	- Operation & feedback of timing pulse sprocket on pusher drive shaft.	✓	○	○
39	Pusher & Cross Conveyor Isolators	- Location of Pusher & Cross conveyor motor isolators.	✓	○	○
40	X-Conveyor Cross-Sensor	- Location & function of Cross Conveyor Clear cross photocell.	✓	○	○
41	Fingers Clear Photocells	- Locations & function of "Outfeed Fingers Clear" photocells.	✓	○	○
42	Finger Tilt Proxies	Location & operation of "Finger Tilt" proxies & safe reset procedure in scenario of "jam-up" using manual pendant.	✓	○	○

37. Unloader Rear Home & Front Push position Proxies





38. Pulse Sprocket



39. Pusher & Cross Conveyor Isolators



40. X-Conveyor Cross-Sensor





41. Fingers Clear Photocells

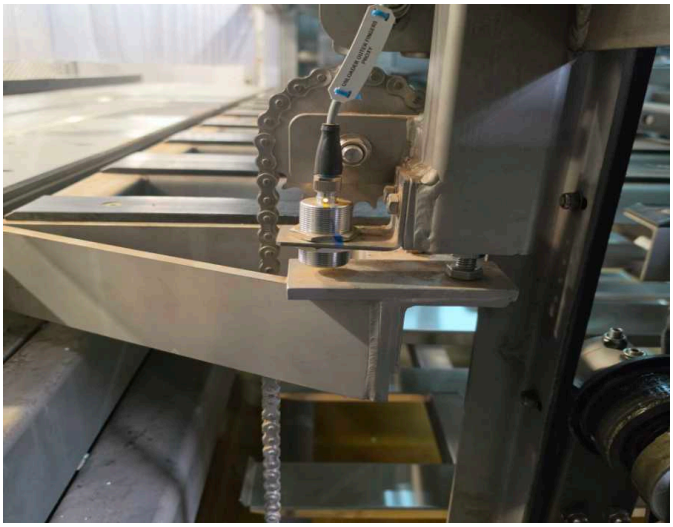




42. Finger Tilt Proxies







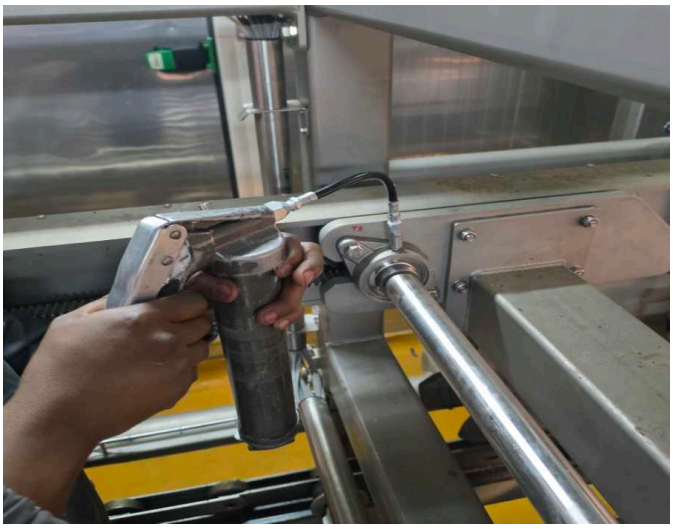
3.2 Prover Unloader Maintenance

SN	Inspection Item	Note	Good	Repair	Replace	NA
43	Gearmotor/Drive Mechanism Assembly	- Pusher chain tensioned. Sprocket tooth inspection. - Cross conveyor bearings greased. - Gearbox oil samples/6-12 monthly oil change. - Noise/vibration monitoring	✓	○	○	○
44	Bearings, Rollers, Rack & Pinion, Drive Coupling	- Grease bearings monthly. - Routine clean & re-grease of rack & pinion. - Roller inspection. - Main drive coupling inspection.	✓	○	○	○
45	Cross Conveyor	- Slat belt & wearstrip inspection. - Sprocket & roller inspection. - Bearing inspection/re-grease.	✓	○	○	○
46	Home & Offload Position Proxies	- Inspection of Home and Offload proxies for tightness & correct positioning, cable tightness.	✓	○	○	○

43. Gearmotor/Drive Mechanism Assembly



44. Bearings, Rollers, Rack & Pinion, Drive Coupling





45. Cross Conveyor





4.0 Safety Mechanisms

SN	Inspection Item	Note	Good	Repair	Replace	NA
47	Fortress Lock Lockout & E-Stop	- Run inhibit of machine when fortress lockout is turned off. - Key access to restricted areas (Loader/Unloader/Access doors)	✓	○	○	○
48	Magveyor Safety Cage	- Run-inhibit of loader when magveyor safety cage is opened. - Safe recovery of possible trigger scenarios.	✓	○	○	○
49	Escape Door Safety Switch	- Routine function test of Escape Door Safety Switch.	✓	○	○	○
50	Infeed Finger Tilt Proxies	- Explanation of function of finger tilt proxies, safe recovery and reset procedure.	✓	○	○	○
51	Outfeed Finger Tilt Proxies	- Explanation of function of finger tilt proxies, safe recovery and reset procedure.	✓	○	○	○
52	Infeed Under/Over Push & Cross Sensor Photocells	- Explanation of Over/Under Push conditions and safe recovery and reset procedure of condition.	✓	○	○	○
53	Infeed Swing Tilt Proxies	- Explanation of swing tilt condition & safe recovery and reset procedure.	✓	○	○	○

SN	Inspection Item	Note	Pass	Fail	NA
54	Panel E-Stop	- Routine function test & reset procedure of Panel E-Stop.	✓	○	○
55	Safe Load/Unload Encoder Positions	- Explanation of "Safe Windows" during prover cycling. - Run inhibit when outside of safe windows, manual recovery procedure using pendant.	✓	○	○

47. Fortress Lock Lockout & E-Stop



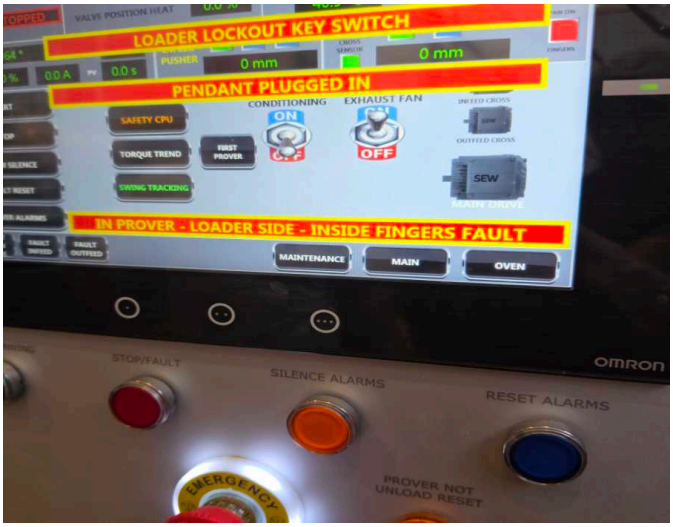
48. Magveyor Safety Cage



50. Escape Door Safety Switch



51. Infeed Finger Tilt Proxies







52. Outfeed Finger Tilt Proxies



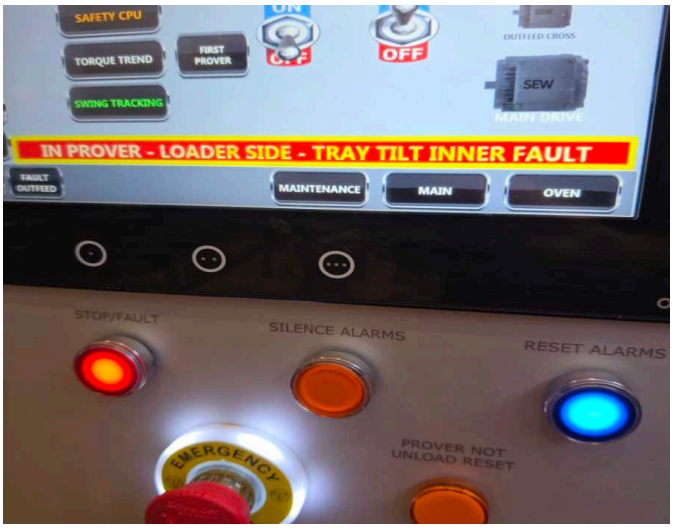


53. Infeed Under/Over Push & Cross Sensor Photocells



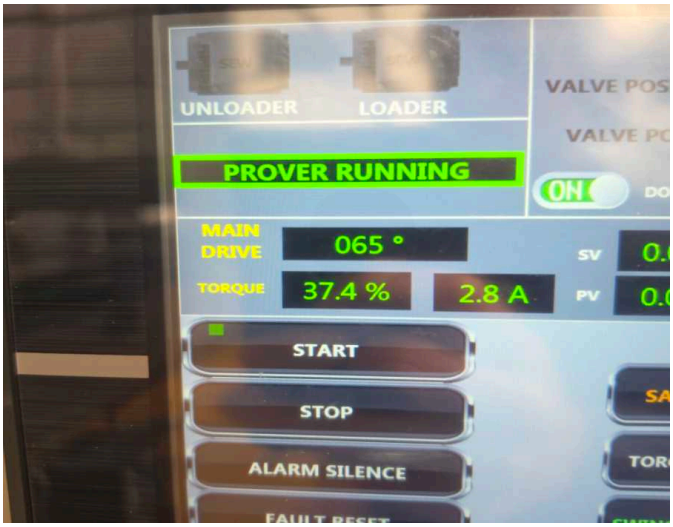


54. Infeed Swing Tilt Proxies





55. Safe Load/Unload Encoder Positions





5.0 Training Register

SN	Inspection Item	Note	Good	Repair	Replace	NA
56	Signed Training Register		✓	○	○	○

SN	Inspection Item	Note	Pass	Fail	NA
57	Trainee 1	Wasfiq Theunissen	✓	○	○
58	Trainee 2	Delmaine Fred	✓	○	○
59	Trainee 3	Tumisho Molefe	✓	○	○
60	Trainee 4	Piet Matjomane	✓	○	○

Additional Note
This document serves as an overview of training provided to the maintenence team and does not serve as the official operation/maintenence manual which will be provided separately.

Overall Condition:	Excellent Condition
Asset Safe to Use:	**Yes**
Maintenance Required:	No
Asset Status	Available

Custom Business Fields

Client Name : Blue Ribbon P1 Maintenance Team

Contact details : 403-4



Dale Technician Signature

Client Signature